

# EE 101: Basic Electronics

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# Sinusoids and Phasors

# Sinusoids

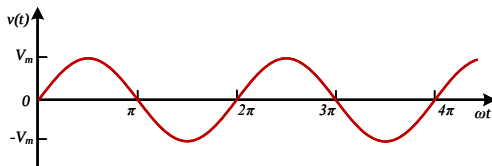
$$v(t) = V_m \sin \omega t$$

$V_m$  = the amplitude of sinusoid

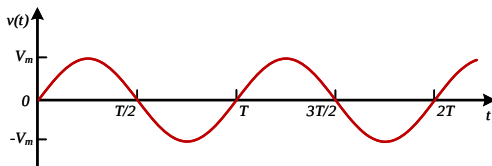
$\omega$  = the angular frequency in radians/s

$\omega t$  = the argument of the sinusoid

$$T = \frac{2\pi}{\omega}$$



(a)



(b)

## Sinusoids contd.

$$\begin{aligned}v(t + T) &= V_m \sin \omega(t + T) = V_m \sin \omega \left( t + \frac{2\pi}{\omega} \right) \\&= V_m \sin(\omega t + 2\pi) = V_m \sin \omega t = v(t)\end{aligned}$$

$$\boxed{v(t + T) = v(t)}$$

- A **periodic function** is one that satisfies  $f(t) = f(t + nT)$  for all  $t$  and for all integers  $n$ .

$$\boxed{f = \frac{1}{T}}$$

$$\omega = 2\pi f$$

# Sinusoids contd.

$$v(t) = V_m \sin(\omega t + \phi)$$

$$v_1(t) = V_m \sin \omega t \text{ and}$$

$$v_2(t) = V_m \sin(\omega t + \phi)$$

- $v_2$  leads  $v_1$  by  $\phi$  or,  $v_1$  lags  $v_2$  by  $\phi$ .

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

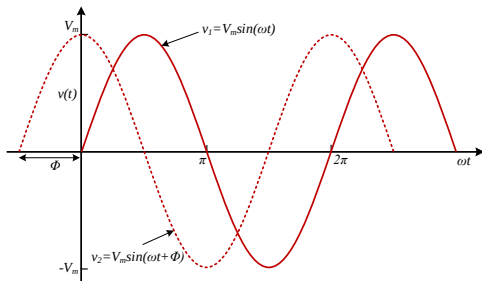
$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(\omega t \pm 180^\circ) = -\sin(\omega t)$$

$$\cos(\omega t \pm 180^\circ) = -\cos(\omega t)$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos(\omega t)$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin(\omega t)$$

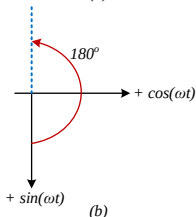
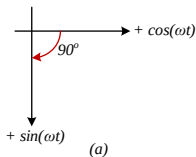


# Sinusoids contd.

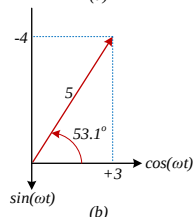
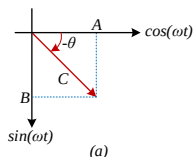
$$A\cos\omega t + B\sin\omega t = C\cos(\omega t - \theta)$$

$$C = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A}$$

$$3\cos\omega t - 4\sin\omega t = 5\cos(\omega t + 53.1^\circ)$$



**Figure:** A graphical means of relating cosine and sine: (a)  $\cos(\omega t - 90^\circ) = \sin(\omega t)$ , (b)  $\sin(\omega t + 180^\circ) = -\sin(\omega t)$ .



**Figure:** (a) Adding  $A\cos\omega t$  and  $B\sin\omega t$ , (b) adding  $3\cos\omega t$  and  $-4\sin\omega t$ .

# Phasors

- A **phasor** is a complex number that represents the amplitude and phase of a sinusoid.

$$\bar{z} = x + jy$$

$$\bar{z} = r\angle\phi = re^{j\phi}$$

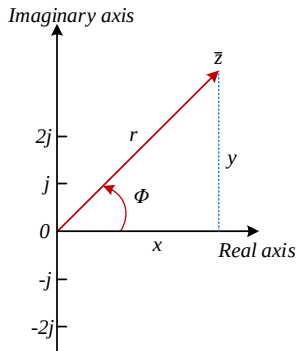
$$\bar{z} = x + jy \quad \text{Rectangular form}$$

$$\bar{z} = r\angle\phi \quad \text{Polar form}$$

$$\bar{z} = re^{j\phi} \quad \text{Exponential form}$$

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}$$

$$x = r\cos\phi, \quad y = r\sin\phi$$



**Figure:** Representation of a complex number  $\bar{z} = x + jy = r\angle\phi$ .

$$\bar{z} = x + jy = r\angle\phi = r(\cos\phi + j\sin\phi)$$

## Phasors contd.

$$\bar{z} = x + jy = r\angle\phi, \quad \bar{z}_1 = x_1 + jy_1 = r_1\angle\phi_1, \quad \bar{z}_2 = x_2 + jy_2 = r_2\angle\phi_2$$

- Addition:  $\bar{z}_1 + \bar{z}_2 = (x_1 + x_2) + j(y_1 + y_2)$
- Subtraction:  $\bar{z}_1 - \bar{z}_2 = (x_1 - x_2) + j(y_1 - y_2)$
- Multiplication:  $\bar{z}_1 \bar{z}_2 = r_1 r_2 \angle \phi_1 + \phi_2$
- Division:  $\frac{\bar{z}_1}{\bar{z}_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$
- Reciprocal:  $\frac{1}{\bar{z}} = \frac{1}{r} \angle -\phi$
- Square root:  $\sqrt{\bar{z}} = \sqrt{r} \angle \phi/2$
- Complex conjugate:  $\bar{z}^* = x - jy = r \angle -\phi = r e^{-j\phi} \quad \left(\frac{1}{j} = -j\right)$



# Phasors contd.

$$e^{\pm j\phi} = \cos\phi \pm j\sin\phi$$

$$\cos\phi = \operatorname{Re}(e^{j\phi})$$

$$\sin\phi = \operatorname{Im}(e^{j\phi})$$

$$\begin{aligned} v(t) &= V_m \cos(\omega t + \phi) \\ &= \operatorname{Re}(V_m e^{j(\omega t + \phi)}) \end{aligned}$$

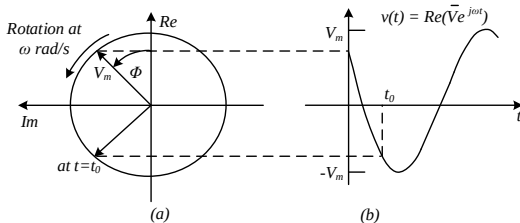
$$v(t) = \operatorname{Re}(V_m e^{j\phi} e^{j\omega t})$$

$$v(t) = \operatorname{Re}(\bar{V} e^{j\omega t})$$

$$\bar{V} = V_m e^{j\phi} = V_m \angle \phi$$

$$v(t) = V_m \cos(\omega t + \phi)$$

$$\bar{V} = V_m \angle \phi$$



**Figure:** Representation of  $\bar{V}e^{j\omega t}$ : (a) sinor rotating counterclockwise, (b) its projection on the real axis, as a function of time.

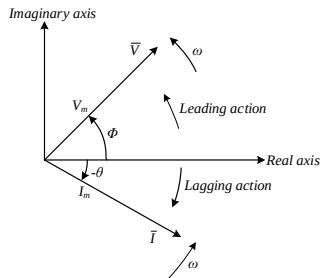
## Phasors contd.

| Time domain representation    | Phasor domain representation   |
|-------------------------------|--------------------------------|
| $V_m \cos(\omega t + \phi)$   | $V_m \angle \phi$              |
| $V_m \sin(\omega t + \phi)$   | $V_m \angle \phi - 90^\circ$   |
| $I_m \cos(\omega t + \theta)$ | $I_m \angle \theta$            |
| $I_m \sin(\omega t + \theta)$ | $I_m \angle \theta - 90^\circ$ |
| $\frac{dv}{dt}$               | $j\omega \bar{V}$              |
| $\int v dt$                   | $\frac{\bar{V}}{j\omega}$      |

$$\begin{aligned} \frac{dv}{dt} &= -\omega V_m \sin(\omega t + \phi) = \omega V_m \cos(\omega t + \phi + 90^\circ) \\ &= \operatorname{Re}(\omega V_m e^{j\omega t} e^{j\phi} e^{j90^\circ}) = \operatorname{Re}(j\omega \bar{V} e^{j\omega t}) \end{aligned}$$

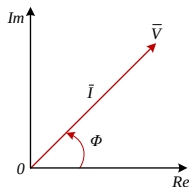
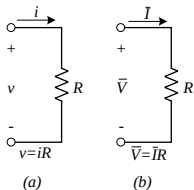
The differences between  $v(t)$  and  $\bar{V}$  should be emphasized:

- $v(t)$  is the *instantaneous or time domain* representation, while  $\bar{V}$  is the *frequency or phasor domain* representation.
- $v(t)$  is time dependent, while  $\bar{V}$  is not. (This fact is often forgotten by students.)
- $v(t)$  is always real with no complex term, while  $\bar{V}$  is generally complex.



**Figure:** A phasor diagram showing  $\bar{V} = V_m \angle \phi$  and  $\bar{I} = I_m \angle -\theta$ .

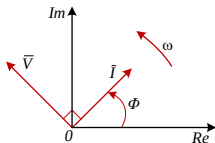
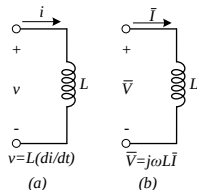
# Phasors and Circuit Elements



$$v = iR = RI_m \cos(\omega t + \phi)$$

$$\bar{V} = RI_m \angle \phi$$

$$\bar{V} = R\bar{I}$$



$$v = L \frac{di}{dt} = -\omega LI_m \sin(\omega t + \phi)$$

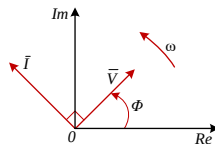
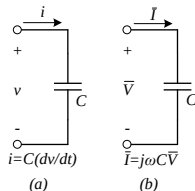
$$v = \omega LI_m \cos(\omega t + \phi + 90^\circ)$$

$$\bar{V} = \omega LI_m e^{j(\phi + 90^\circ)}$$

$$= \omega LI_m e^{j\phi} e^{j90^\circ}$$

$$= \omega LI_m \angle \phi + 90^\circ$$

$$\bar{V} = j\omega L\bar{I}$$



$$i = C \frac{dv}{dt}$$

$$\bar{I} = j\omega C\bar{V} \Rightarrow \bar{V} = \frac{\bar{I}}{j\omega C}$$

## Phasors and Circuit Elements contd.

**Table:** Summary of Voltage-Current Relationships

| Element | Time domain           | Frequency domain                      |
|---------|-----------------------|---------------------------------------|
| $R$     | $v = Ri$              | $\bar{V} = R\bar{I}$                  |
| $L$     | $v = L \frac{di}{dt}$ | $\bar{V} = j\omega L\bar{I}$          |
| $C$     | $i = C \frac{dv}{dt}$ | $\bar{V} = \frac{\bar{I}}{j\omega C}$ |

# Impedance and Admittance

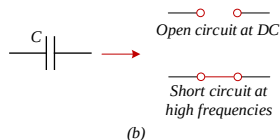
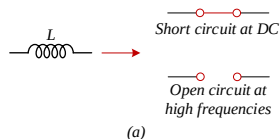
$$\bar{V} = R\bar{I}, \quad \bar{V} = j\omega L\bar{I}, \quad \bar{V} = \frac{\bar{I}}{j\omega C}$$

$$\frac{\bar{V}}{\bar{I}} = R, \quad \frac{\bar{V}}{\bar{I}} = j\omega L, \quad \frac{\bar{V}}{\bar{I}} = \frac{1}{j\omega C}$$

$$\boxed{\bar{Z} = \frac{\bar{V}}{\bar{I}} \quad \text{or,} \quad \bar{V} = \bar{Z}\bar{I}}$$

- The **impedance**  $\bar{Z}$  of a circuit is the ratio of the phasor voltage  $\bar{V}$  to the phasor current  $\bar{I}$ , measured in ohms ( $\Omega$ ).

| Element | Impedance                       | Admittance                      |
|---------|---------------------------------|---------------------------------|
| $R$     | $\bar{Z} = R$                   | $\bar{Y} = \frac{1}{R}$         |
| $L$     | $\bar{Z} = j\omega L$           | $\bar{Y} = \frac{1}{j\omega L}$ |
| $C$     | $\bar{Z} = \frac{1}{j\omega C}$ | $\bar{Y} = j\omega C$           |



## Impedance and Admittance contd.

$$\bar{Z} = R + jX$$

$$\bar{Z} = |\bar{Z}| \angle \theta$$

$$\boxed{\bar{Z} = R + jX = |\bar{Z}| \angle \theta}$$

$$|\bar{Z}| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R}$$

$$R = |\bar{Z}| \cos \theta, \quad X = |\bar{Z}| \sin \theta$$

- The **admittance**  $\bar{Y}$  is the reciprocal of impedance, measured in siemens ( $S$ ).

$$\boxed{\bar{Y} = \frac{1}{\bar{Z}} = \frac{\bar{I}}{\bar{V}}}$$

$$\boxed{\bar{Y} = G + jB}$$

- $G = \operatorname{Re} \bar{Y}$  is called the *conductance*.
- $B = \operatorname{Im} \bar{Y}$  is called the *susceptance*.

# Kirchhoff's Laws in Frequency Domain

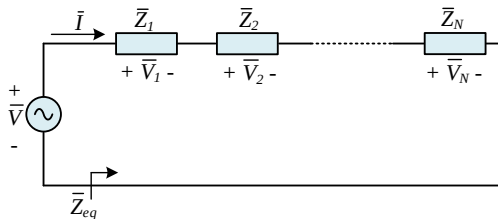
$$\begin{aligned}v_1 + v_2 + \cdots + v_n &= 0 \\V_{m1}\cos(\omega t + \theta_1) + V_{m2}\cos(\omega t + \theta_2) \\&+ \cdots + V_{mn}\cos(\omega t + \theta_n) = 0\end{aligned}$$

$$\begin{aligned}\operatorname{Re}(V_{m1}e^{j\theta_1}e^{j\omega t}) + \operatorname{Re}(V_{m2}e^{j\theta_2}e^{j\omega t}) + \cdots + \operatorname{Re}(V_{mn}e^{j\theta_n}e^{j\omega t}) &= 0 \\ \operatorname{Re}\left[(V_{m1}e^{j\theta_1} + V_{m2}e^{j\theta_2} + \cdots + V_{mn}e^{j\theta_n})e^{j\omega t}\right] &= 0\end{aligned}$$

$$\begin{aligned}\operatorname{Re}\left[(\bar{V}_1 + \bar{V}_2 + \cdots + \bar{V}_n)e^{j\omega t}\right] &= 0 \\ \bar{V}_1 + \bar{V}_2 + \cdots + \bar{V}_n &= 0\end{aligned}$$

$$\begin{aligned}i_1 + i_2 + \cdots + i_n &= 0 \\ \bar{I}_1 + \bar{I}_2 + \cdots + \bar{I}_n &= 0\end{aligned}$$

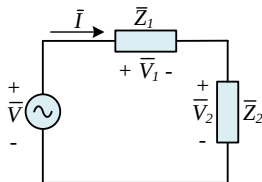
# Impedance Combinations



$$\bar{V} = \bar{V}_1 + \bar{V}_2 + \dots + \bar{V}_N = \bar{I}(\bar{Z}_1 + \bar{Z}_2 + \dots + \bar{Z}_N)$$

$$\bar{Z}_{eq} = \frac{\bar{V}}{\bar{I}} = \bar{Z}_1 + \bar{Z}_2 + \dots + \bar{Z}_N$$

$$\boxed{\bar{Z}_{eq} = \bar{Z}_1 + \bar{Z}_2 + \dots + \bar{Z}_N}$$

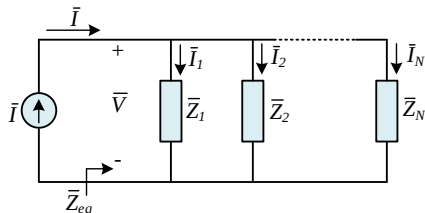


$$\bar{I} = \frac{\bar{V}}{\bar{Z}_1 + \bar{Z}_2}$$

$$\bar{V}_1 = \frac{\bar{Z}_1}{\bar{Z}_1 + \bar{Z}_2} \bar{V}, \quad \bar{V}_2 = \frac{\bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2} \bar{V}$$



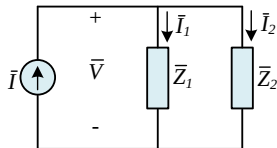
# Impedance Combinations contd.



$$\begin{aligned}\bar{I} &= \bar{I}_1 + \bar{I}_2 + \dots + \bar{I}_N \\ &= \bar{V} \left( \frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} + \dots + \frac{1}{\bar{Z}_N} \right)\end{aligned}$$

$$\frac{1}{\bar{Z}_{eq}} = \frac{\bar{I}}{\bar{V}} = \frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} + \dots + \frac{1}{\bar{Z}_N}$$

$$\boxed{\bar{Y}_{eq} = \bar{Y}_1 + \bar{Y}_2 + \dots + \bar{Y}_N}$$



$$\begin{aligned}\bar{Z}_{eq} &= \frac{1}{\bar{Y}_{eq}} = \frac{1}{\bar{Y}_1 + \bar{Y}_2} \\ &= \frac{1}{1/\bar{Z}_1 + 1/\bar{Z}_2} = \frac{\bar{Z}_1 \bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2}\end{aligned}$$

$$\bar{V} = \bar{I} \bar{Z}_{eq} = \bar{I}_1 \bar{Z}_1 = \bar{I}_2 \bar{Z}_2$$

$$\bar{I}_1 = \frac{\bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2} \bar{I}, \quad \bar{I}_2 = \frac{\bar{Z}_1}{\bar{Z}_1 + \bar{Z}_2} \bar{I}$$

Thank You!